

## EC2 instances types.

### 1. General Purpose Instance

These provide a balanced mix of CPU, memory, and networking.

Examples

- M series (M5, M6i)
- T series (T2, T3, T4g)
- A1

Use cases

- Web servers
- Small databases
- Dev/test environments
- Microservices

Reality:

Most beginners use T2 or T3 because they are cheap and burstable.

### 2. Compute Optimized Instances

These provide high CPU power.

Examples

- C5
- C6i
- C7g

Use cases

- High performance computing
- Gaming servers

- Batch processing
- Media transcoding

If your application is CPU-heavy, using general instances is inefficient.

### 3. Memory Optimized Instances

These provide large RAM capacity.

Examples

- R5 / R6i
- X1 / X2
- z1d

Use cases

- Large databases
- In-memory caching
- Real-time big data analytics

If the application constantly loads large datasets, you need this.

### 4. Storage Optimized Instances

Designed for high disk throughput and IOPS.

Examples

- I3 / I4
- D2
- H1

Use cases

- Big data processing

- Data warehouses
- Log processing systems

## 5. Accelerated Computing Instances

Uses GPU or FPGA hardware acceleration.

Examples

- P series (machine learning training)
- G series (graphics / video rendering)
- F series (FPGA)

Use cases

- Machine learning
- AI model training
- Video rendering
- 3D graphics

## AWS CLI connection

### 1. What is AWS CLI

The AWS Command Line Interface (CLI) is a command-line tool that enables users to interact with AWS services through commands.

Using AWS CLI, users can create, manage, and monitor resources such as:

- Amazon EC2 instances
- Amazon S3 buckets
- Amazon Elastic Kubernetes Service clusters
- AWS Identity and Access Management users and roles

It is widely used in automation, scripting, and DevOps workflows.

## 2. Purpose of AWS CLI Connection

The main purpose of connecting AWS CLI to an AWS account is to:

- Manage AWS services from a terminal
- Automate infrastructure operations
- Integrate with CI/CD pipelines
- Execute AWS commands quickly without logging into the AWS console

This improves efficiency and allows infrastructure to be managed programmatically.

## 3. Authentication Process

AWS CLI connection requires authentication using security credentials.

The authentication is done through:

- Access Key ID
- Secret Access Key
- Region configuration
- Output format

This is the creation of the EC2 instance using the CLI.

```
aws ec2 run-instances \  
--image-id <ami-xxxxxxx> \  
--instance-type <type of instances> \  
--key-name <key name> \  
--security-group-ids <security group name> \  
--count 1
```

## Creation of ec2 instance.

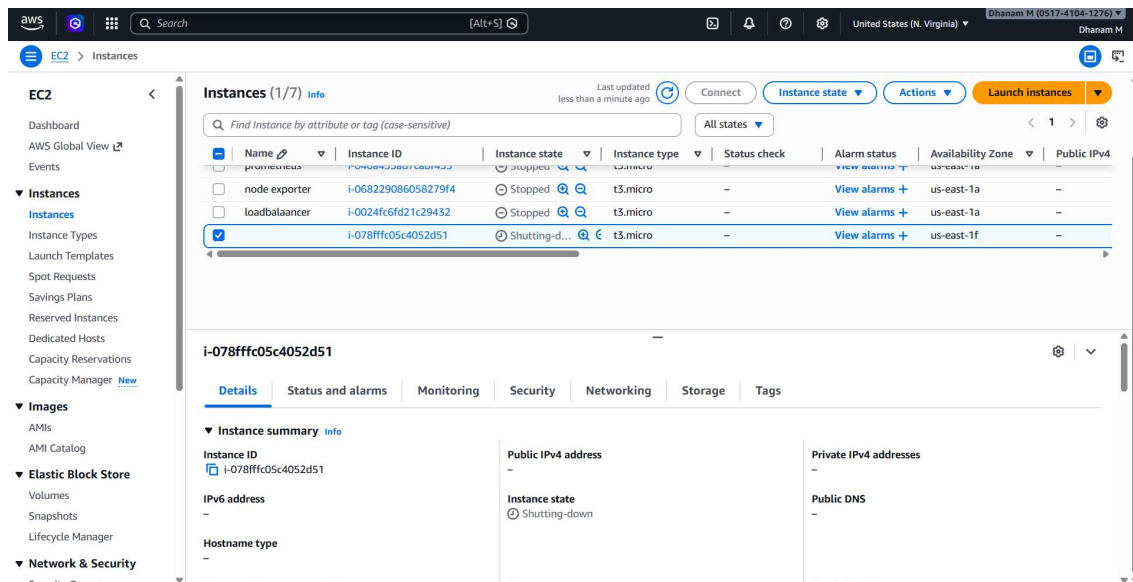
```
hariharan@hari-3K39PR78 MINGW64 -
$ aws ec2 run-instances --image-id ami-0f3caalc4417e51b --instance-type t3.micro --key-name mykey --security-group-ids "sg-0a91bfc93233c9191" --count 1
{
  "ReservationId": "r-0f87bf622e8f925c3",
  "OwnerId": "051741041276",
  "Groups": [],
  "Instances": [
    {
      "Architecture": "x86_64",
      "BlockDeviceMappings": [],
      "ClientToken": "578513f6-b7fe-4e71-b583-e5194271d1f8",
      "EbsOptimized": false,
      "EnaSupport": true,
      "Hypervisor": "xen",
      "NetworkInterfaces": [
        {
          "Attachment": {
            "AttachTime": "2026-03-06T07:02:28+00:00",
            "AttachmentId": "eni-attach-02bc30b2c52815454",
            "DeleteOnTermination": true,
            "DeviceIndex": 0,
            "Status": "attaching",
            "NetworkCardIndex": 0
          },
          "Description": "",
          "Groups": [
            {
              "GroupId": "sg-0a91bfc93233c9191",
              "GroupName": "default"
            }
          ],
          "Ipv6Addresses": [],
          "MacAddress": "16:ff:df:41:ee:2d",
          "NetworkInterfaceId": "eni-0dc4d4340cf209895",
          "OwnerId": "051741041276",
          "PrivateDnsName": "ip-172-31-74-66.ec2.internal",
          "PrivateIpAddress": "172.31.74.66",
          "PrivateIpAddresses": [
            {
              "Primary": true,
              "PrivateDnsName": "ip-172-31-74-66.ec2.internal",
              "PrivateIpAddress": "172.31.74.66"
            }
          ],
          "SourceDestCheck": true,
          "Status": "in-use",
          "SubnetId": "subnet-07e40a724fbc9e7ed",
          "VpcId": "vpc-0be33b50250021280"
        }
      ]
    }
  ]
}
```

The screenshot shows the AWS Management Console for the 'Instances' page. The top navigation bar includes the AWS logo, a search bar, and the user's profile 'Dhanam M (0517-4104-1276)'. The left sidebar contains a navigation menu with options like 'Dashboard', 'Events', 'Instances', 'Instance Types', 'Launch Templates', 'Spot Requests', 'Savings Plans', 'Reserved Instances', 'Dedicated Hosts', 'Capacity Reservations', 'Capacity Manager', 'Images', 'AMI Catalog', 'Elastic Block Store', 'Volumes', 'Snapshots', 'Lifecycle Manager', and 'Network & Security'. The main content area shows a table of instances with columns for Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, and Public IPv4. The table lists three instances: 'node exporter' (Stopped), 'loadbalancer' (Stopped), and 'i-078fffc05c4052d51' (Running). Below the table, there is a 'Select an instance' section.

| Name                | Instance ID         | Instance state | Instance type | Status check | Alarm status  | Availability Zone | Public IPv4 |
|---------------------|---------------------|----------------|---------------|--------------|---------------|-------------------|-------------|
| node exporter       | i-068229086058279f4 | Stopped        | t3.micro      | -            | View alarms + | us-east-1a        | -           |
| loadbalancer        | i-0024fc6fd1c29432  | Stopped        | t3.micro      | -            | View alarms + | us-east-1a        | -           |
| i-078fffc05c4052d51 | i-078fffc05c4052d51 | Running        | t3.micro      | Initializing | View alarms + | us-east-1f        | ec2-3-230-1 |

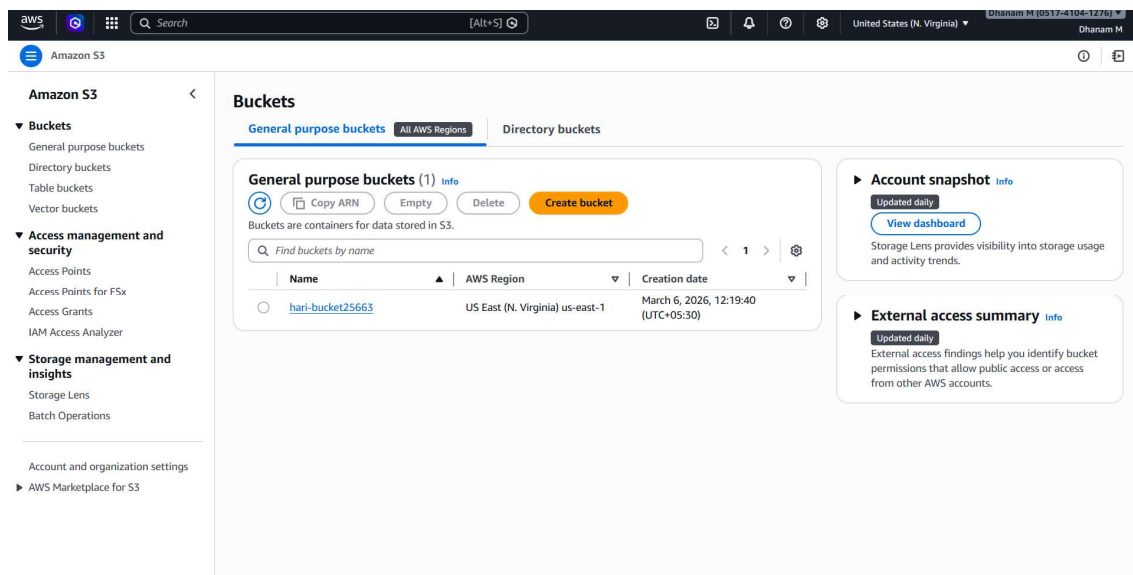
## Deletion of EC2 instances.

```
hariharan@hari-3K39PR78 MINGW64 -
$ aws ec2 terminate-instances --instance-ids i-078fffc05c4052d51
{
  "TerminatingInstances": [
    {
      "InstanceId": "i-078fffc05c4052d51",
      "CurrentState": {
        "Code": 32,
        "Name": "shutting-down"
      },
      "PreviousState": {
        "Code": 16,
        "Name": "running"
      }
    }
  ]
}
```



## Creation of S3 Bucket.

```
hariharan@hari-JK39PR78 MINGW64 ~
$ aws s3 mb s3://hari-bucket25663
make_bucket: hari-bucket25663
hariharan@hari-JK39PR78 MINGW64 ~
$
```



## And the removal of S3 Bucket.

```
hariharan@hari-JK39PR78 MINGW64 ~
$ aws s3 rb s3://hari-bucket25663
remove_bucket: hari-bucket25663
hariharan@hari-JK39PR78 MINGW64 ~
$
```